HUFT – E-COMMERCE ORDER MANAGEMENT DATABASE SYSTEM

WEEK 8 – DATABASE RELATIONSHIP ANALYSIS USING JOINS

Task 1 – INNER JOIN: Customer and Orders

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  6      o.Total_Amount,
  7      o.Order_Status
  8  FROM CUST3 c
  9  INNER JOIN HuftOrder o
 10      ON c.CID = o.Customer_ID
 11  ORDER BY c.CID;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
----------- -------------------- -------- ------------ ------------ ---------------
          1 Abi                      1001 18-SEP-2026     1,750.00 Delivered
          2 Anu                      1002 18-SEP-2026     1,849.00 Shipped
          3 Ravi                     1003 19-SEP-2026     1,099.00 Processing
          4 Karthik                  1004 19-SEP-2026     1,273.00 Delivered
          5 Priya                    1005 20-SEP-2026     1,075.00 Shipped
          6 Divya                    1006 20-SEP-2026       650.00 Processing
          7 Arun                     1007 20-SEP-2026     1,197.00 Delivered
          8 Meena                    1008 21-SEP-2026     1,498.00 Processing
         10 Nisha                    1010 21-SEP-2026       900.00 Processing

9 rows selected.
```

Task 2 – INNER JOIN: Orders and Payment

```
SQL> SELECT
  2      o.Order_ID,
  3      o.Order_Date,
  4      o.Total_Amount,
  5      o.Order_Status,
  6      p.Payment_ID,
  7      p.Payment_Method,
  8      p.Payment_Amount,
  9      p.Payment_Status
 10  FROM HuftOrder o
 11  LEFT JOIN HuftPayment p
 12      ON o.Order_ID = p.Order_ID
 13  ORDER BY o.Order_ID;

ORDER_ID ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS    PAYMENT_ID PAYMENT_METHOD       PAYMENT_AMOUNT PAYMENT_STATUS
-------- ------------ ------------ --------------- ---------- -------------------- -------------- ---------------
    1001 18-SEP-26       1,750.00 Delivered                1 UPI                          1750 Paid
    1002 18-SEP-26       1,849.00 Shipped                  2 Credit Card                  1849 Paid
    1003 19-SEP-26       1,099.00 Processing               3 Debit Card                   1099 Paid
    1004 19-SEP-26       1,273.00 Delivered                4 UPI                          1273 Paid
    1005 20-SEP-26       1,075.00 Shipped                  5 Cash on Delivery             1075 Pending
    1006 20-SEP-26         650.00 Processing               6 UPI                           650 Paid
    1007 20-SEP-26       1,197.00 Delivered                7 Net Banking                  1197 Paid
    1008 21-SEP-26       1,498.00 Processing               8 Credit Card                  1498 Paid
    1009 21-SEP-26         998.00 Shipped                  9 UPI                           998 Paid
    1010 21-SEP-26         900.00 Processing              10 Cash on Delivery              900 Pending

10 rows selected.
```

Task 3 – INNER JOIN: Product and Order Details

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      oi.Order_ID,
  5      oi.Quantity,
  6      oi.Price,
  7      oi.Total_Amount
  8  FROM HuftProduct p
  9  INNER JOIN HuftOrder_Item oi
 10      ON p.Product_ID = oi.Product_ID
 11  ORDER BY p.Product_ID;

PRODUCT_ID PRODUCT_NAME                  ORDER_ID QUANTITY       PRICE TOTAL_AMOUNT
---------- ----------------------------- -------- -------- ----------- ------------
       101 Chicken Dog Food                  1001        2      650.00     1,300.00
       101 Chicken Dog Food                  1006        1      650.00       650.00
       102 Grain Free Cat Food               1002        1      749.00       749.00
       102 Grain Free Cat Food               1008        2      749.00     1,498.00
       103 Leather Dog Collar                1001        1      450.00       450.00
       103 Leather Dog Collar                1010        2      450.00       900.00
       104 Pet Grooming Shampoo              1003        2      350.00       700.00
       106 Cat Litter                        1002        2      550.00     1,100.00
       107 Dental Chew Sticks                1003        1      399.00       399.00
       107 Dental Chew Sticks                1007        3      399.00     1,197.00
       108 Pet Feeding Bowl                  1004        2      275.00       550.00
       109 Puppy Training Pads               1004        1      499.00       499.00
       109 Puppy Training Pads               1009        2      499.00       998.00
       110 Pet Conditioner                   1005        1      425.00       425.00
       112 Cat Grooming Brush                1005        2      325.00       650.00

15 rows selected.
```

Task 4 – LEFT JOIN: Display All Customers and Their Orders

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount,
  7      o.Order_Status
  8  FROM CUST3 c
  9  LEFT JOIN HuftOrder o
 10      ON c.CID = o.Customer_ID
 11  ORDER BY c.CID;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
----------- -------------------- -------- ----------- ------------ ---------------
          1 Abi                     1001 18-SEP-26       1,750.00 Delivered
          2 Anu                     1002 18-SEP-26       1,849.00 Shipped
          3 Ravi                    1003 19-SEP-26       1,099.00 Processing
          4 Karthik                 1004 19-SEP-26       1,273.00 Delivered
          5 Priya                   1005 20-SEP-26       1,075.00 Shipped
          6 Divya                   1006 20-SEP-26         650.00 Processing
          7 Arun                    1007 20-SEP-26       1,197.00 Delivered
          8 Meena                   1008 21-SEP-26       1,498.00 Processing
         10 Nisha                   1010 21-SEP-26         900.00 Processing

9 rows selected.
```

Task 5 – RIGHT JOIN: Display All Orders and Customer Details

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount,
  7      o.Order_Status
  8  FROM CUST3 c
  9  RIGHT JOIN HuftOrder o
 10      ON c.CID = o.Customer_ID
 11  ORDER BY o.Order_ID;

CUSTOMER_ID CUSTOMER_NAME        ORDER_ID ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
----------- -------------------- -------- ----------- ------------ ---------------
          1 Abi                      1001 18-SEP-26       1,750.00 Delivered
          2 Anu                      1002 18-SEP-26       1,849.00 Shipped
          3 Ravi                     1003 19-SEP-26       1,099.00 Processing
          4 Karthik                  1004 19-SEP-26       1,273.00 Delivered
          5 Priya                    1005 20-SEP-26       1,075.00 Shipped
          6 Divya                    1006 20-SEP-26         650.00 Processing
          7 Arun                     1007 20-SEP-26       1,197.00 Delivered
          8 Meena                    1008 21-SEP-26       1,498.00 Processing
                                     1009 21-SEP-26         998.00 Shipped
         10 Nisha                    1010 21-SEP-26         900.00 Processing

10 rows selected.
```

Task 6 – Complete Order Details

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      p.Product_Name,
  6      oi.Quantity,
  7      oi.Price,
  8      oi.Total_Amount AS Item_Amount,
  9      o.Order_Status,
 10      pay.Payment_Method,
 11      pay.Payment_Status
 12  FROM CUST3 c
 13  JOIN HuftOrder o
 14      ON c.CID = o.Customer_ID
 15  JOIN HuftOrder_Item oi
 16      ON o.Order_ID = oi.Order_ID
 17  JOIN HuftProduct p
 18      ON oi.Product_ID = p.Product_ID
 19  LEFT JOIN HuftPayment pay
 20      ON o.Order_ID = pay.Order_ID
 21  ORDER BY o.Order_ID;
```

```
CUSTOMER_ID CUSTOMER_NAME      ORDER_ID PRODUCT_NAME             QUANTITY      PRICE ITEM_AMOUNT ORDER_STATUS PAYMENT_METHOD     PAYMENT_
STAT
----------- ----------------- -------- ------------------------ -------- ---------- ----------- ------------ ----------------- --------
----
          1 Abi                  1001 Chicken Dog Food                2     650.00        1300 Delivered    UPI               Paid
          1 Abi                  1001 Leather Dog Collar              1     450.00         450 Delivered    UPI               Paid
          2 Anu                  1002 Cat Litter                      2     550.00        1100 Shipped      Credit Card       Paid
          2 Anu                  1002 Grain Free Cat Food             1     749.00         749 Shipped      Credit Card       Paid
          3 Ravi                 1003 Pet Grooming Shampoo            2     350.00         700 Processing   Debit Card        Paid
          3 Ravi                 1003 Dental Chew Sticks              1     399.00         399 Processing   Debit Card        Paid
          4 Karthik              1004 Pet Feeding Bowl                2     275.00         550 Delivered    UPI               Paid
          4 Karthik              1004 Puppy Training Pads             1     499.00         499 Delivered    UPI               Paid
          5 Priya                1005 Cat Grooming Brush              2     325.00         650 Shipped      Cash on Delivery  Pending
          5 Priya                1005 Pet Conditioner                 1     425.00         425 Shipped      Cash on Delivery  Pending
          6 Divya                1006 Chicken Dog Food                1     650.00         650 Processing   UPI               Paid
          7 Arun                 1007 Dental Chew Sticks              3     399.00        1197 Delivered    Net Banking       Paid
          8 Meena                1008 Grain Free Cat Food             2     749.00        1498 Processing   Credit Card       Paid
         10 Nisha                1010 Leather Dog Collar              2     450.00         900 Processing   Cash on Delivery  Pending

14 rows selected.
```

Task 7: Customer Purchase History

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      p.Product_Name,
  6      oi.Quantity,
  7      oi.Price,
  8      oi.Total_Amount AS Item_Amount,
  9      TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date
 10  FROM CUST3 c
 11  JOIN HuftOrder o
 12      ON c.CID = o.Customer_ID
 13  JOIN HuftOrder_Item oi
 14      ON o.Order_ID = oi.Order_ID
 15  JOIN HuftProduct p
 16      ON oi.Product_ID = p.Product_ID
 17  ORDER BY c.CID, o.Order_Date;
```

```
CUSTOMER_ID CUSTOMER_NAME      ORDER_ID PRODUCT_NAME             QUANTITY      PRICE ITEM_AMOUNT ORDER_DATE
----------- ----------------- -------- ------------------------ -------- ---------- ----------- ------------
          1 Abi                  1001 Chicken Dog Food                2     650.00        1300 18-SEP-2026
          1 Abi                  1001 Leather Dog Collar              1     450.00         450 18-SEP-2026
          2 Anu                  1002 Cat Litter                      2     550.00        1100 18-SEP-2026
          2 Anu                  1002 Grain Free Cat Food             1     749.00         749 18-SEP-2026
          3 Ravi                 1003 Pet Grooming Shampoo            2     350.00         700 19-SEP-2026
          3 Ravi                 1003 Dental Chew Sticks              1     399.00         399 19-SEP-2026
          4 Karthik              1004 Pet Feeding Bowl                2     275.00         550 19-SEP-2026
          4 Karthik              1004 Puppy Training Pads             1     499.00         499 19-SEP-2026
          5 Priya                1005 Cat Grooming Brush              2     325.00         650 20-SEP-2026
          5 Priya                1005 Pet Conditioner                 1     425.00         425 20-SEP-2026
          6 Divya                1006 Chicken Dog Food                1     650.00         650 20-SEP-2026
          7 Arun                 1007 Dental Chew Sticks              3     399.00        1197 20-SEP-2026
          8 Meena                1008 Grain Free Cat Food             2     749.00        1498 21-SEP-2026
         10 Nisha                1010 Leather Dog Collar              2     450.00         900 21-SEP-2026

14 rows selected.
```

Task 8: Customer Purchase History with Payment

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  6      p.Product_Name,
  7      oi.Quantity,
  8      oi.Total_Amount AS Item_Amount,
  9      pay.Payment_Method,
 10      pay.Payment_Amount,
 11      pay.Payment_Status
 12  FROM CUST3 c
 13  JOIN HuftOrder o
 14      ON c.CID = o.Customer_ID
 15  JOIN HuftOrder_Item oi
 16      ON o.Order_ID = oi.Order_ID
 17  JOIN HuftProduct p
 18      ON oi.Product_ID = p.Product_ID
 19  LEFT JOIN HuftPayment pay
 20      ON o.Order_ID = pay.Order_ID
 21  ORDER BY c.CID, o.Order_ID;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DATE	PRODUCT_NAME	QUANTITY	ITEM_AMOUNT	PAYMENT_METHOD	PAYMENT_AMOUNT	PAYMENT_STAT
1	Abi	1001	18-SEP-2026	Chicken Dog Food	2	1,300.00	UPI	1,750.00	Paid
1	Abi	1001	18-SEP-2026	Leather Dog Collar	1	450.00	UPI	1,750.00	Paid
2	Anu	1002	18-SEP-2026	Cat Litter	2	1,100.00	Credit Card	1,849.00	Paid
2	Anu	1002	18-SEP-2026	Grain Free Cat Food	1	749.00	Credit Card	1,849.00	Paid
3	Ravi	1003	19-SEP-2026	Pet Grooming Shampoo	2	700.00	Debit Card	1,099.00	Paid
3	Ravi	1003	19-SEP-2026	Dental Chew Sticks	1	399.00	Debit Card	1,099.00	Paid
4	Karthik	1004	19-SEP-2026	Pet Feeding Bowl	2	550.00	UPI	1,273.00	Paid
4	Karthik	1004	19-SEP-2026	Puppy Training Pads	1	499.00	UPI	1,273.00	Paid
5	Priya	1005	20-SEP-2026	Cat Grooming Brush	2	650.00	Cash on Delivery	1,075.00	Pending
5	Priya	1005	20-SEP-2026	Pet Conditioner	1	425.00	Cash on Delivery	1,075.00	Pending
6	Divya	1006	20-SEP-2026	Chicken Dog Food	1	650.00	UPI	650.00	Paid
7	Arun	1007	20-SEP-2026	Dental Chew Sticks	3	1,197.00	Net Banking	1,197.00	Paid
8	Meena	1008	21-SEP-2026	Grain Free Cat Food	2	1,498.00	Credit Card	1,498.00	Paid
10	Nisha	1010	21-SEP-2026	Leather Dog Collar	2	900.00	Cash on Delivery	900.00	Pending

14 rows selected.

Task 9: Multi-Table Business Report

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      TO_CHAR(o.Order_Date, 'DD-MON-YYYY') AS Order_Date,
  6      p.Product_Name,
  7      oi.Quantity,
  8      oi.Total_Amount AS Item_Amount,
  9      pay.Payment_Method,
 10      pay.Payment_Status
 11  FROM CUST3 c
 12  JOIN HuftOrder o
 13      ON c.CID = o.Customer_ID
 14  JOIN HuftOrder_Item oi
 15      ON o.Order_ID = oi.Order_ID
 16  JOIN HuftProduct p
 17      ON oi.Product_ID = p.Product_ID
 18  LEFT JOIN HuftPayment pay
 19      ON o.Order_ID = pay.Order_ID
 20  ORDER BY c.CID, o.Order_ID;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DATE	PRODUCT_NAME	QUANTITY	ITEM_AMOUNT	PAYMENT_METHOD	PAYMENT_STAT
1	Abi	1001	18-SEP-2026	Chicken Dog Food	2	1,300.00	UPI	Paid
1	Abi	1001	18-SEP-2026	Leather Dog Collar	1	450.00	UPI	Paid
2	Anu	1002	18-SEP-2026	Cat Litter	2	1,100.00	Credit Card	Paid
2	Anu	1002	18-SEP-2026	Grain Free Cat Food	1	749.00	Credit Card	Paid
3	Ravi	1003	19-SEP-2026	Pet Grooming Shampoo	2	700.00	Debit Card	Paid
3	Ravi	1003	19-SEP-2026	Dental Chew Sticks	1	399.00	Debit Card	Paid
4	Karthik	1004	19-SEP-2026	Pet Feeding Bowl	2	550.00	UPI	Paid
4	Karthik	1004	19-SEP-2026	Puppy Training Pads	1	499.00	UPI	Paid
5	Priya	1005	20-SEP-2026	Cat Grooming Brush	2	650.00	Cash on Delivery	Pending
5	Priya	1005	20-SEP-2026	Pet Conditioner	1	425.00	Cash on Delivery	Pending
6	Divya	1006	20-SEP-2026	Chicken Dog Food	1	650.00	UPI	Paid
7	Arun	1007	20-SEP-2026	Dental Chew Sticks	3	1,197.00	Net Banking	Paid
8	Meena	1008	21-SEP-2026	Grain Free Cat Food	2	1,498.00	Credit Card	Paid
10	Nisha	1010	21-SEP-2026	Leather Dog Collar	2	900.00	Cash on Delivery	Pending

14 rows selected.